

FE-DETRAC: An Infrastructure Fisheye Video Benchmark with Distortion-Aware Hybrid Data Association for Multi-Object Tracking

Chung-I Huang, *Member, IEEE*, Wei-Yu Chen, Jun-Wei Hsieh, *Senior Member, IEEE*, Ping-Yang Chen, Ming-Ching Chang, *Senior Member, IEEE*, Munkhjargal Gochoo, *Member, IEEE*, Chun-Ming Tsai, Hsiang-Wei Huang, Vincent S. Tseng, *Fellow, IEEE*, and Jenq-Neng Hwang, *Fellow, IEEE*

Abstract—Wide-angle fisheye cameras are increasingly deployed as infrastructure sensors at urban intersections because a single device can cover multiple lanes and approaches, significantly reducing hardware and installation costs. However, the inherent fisheye projection introduces strong radial distortion, scale shrinkage near the periphery, and frequent occlusions. These characteristics severely degrade the performance of standard detection models and, more critically, disrupt data association in multi-object tracking (MOT) where perspective-only linear motion models are typically assumed.

This paper addresses these challenges by presenting FE-DETRAC, a comprehensive infrastructure fisheye video benchmark for urban traffic detection and tracking. The released benchmark contains 20,000 frames annotated with over 470,000 bounding boxes across 5 road-user categories (pedestrian, bike, car, bus, and truck), captured by 22 fixed fisheye IoT cameras at real intersections in Hsinchu City, Taiwan. The dataset is provided in multiple annotation formats (Pascal-VOC, COCO, MOT, and YOLO) with camera-disjoint splits to enable rigorous, reproducible evaluation of cross-camera generalization.

To address distortion-induced association errors without computationally expensive fisheye dewarping, we propose a Hybrid Data Association (HDA) module. HDA couples a fisheye-view state with a virtual distortion-corrected-view state and applies a dual-view IoU matching score for robust tracklet association. Extensive experiments on FE-DETRAC show that HDA consistently improves strong MOT baselines (e.g., StrongSORT, BoT-SORT-R, OC-SORT, and Deep OC-SORT) by 3–5 HOTA points while retaining real-time feasibility (end-to-end latency < 22 ms per frame on a single RTX 3080). Comprehensive

analyses including (i) α and R sensitivity sweeps, (ii) per-zone periphery evaluation, (iii) cross-dataset validation on FishEye8K, (iv) calibration-error robustness, and (v) statistical significance testing with 95% confidence intervals over 5 runs support the interpretation that the gains are not limited to a single setting. Additional diagnostic analyses report the worst-case Jacobian condition number and corrected-domain covariance eigenvalue spread in peripheral zones, together with a controlled HDA-only latency profile for dense association settings up to $N=400$ candidates per frame. The dataset is publicly released at DOI: 10.30193/scidm-ds-571593m to facilitate future research in fisheye video analytics.

Index Terms—Fisheye cameras, infrastructure surveillance, multi-object tracking, distortion-aware data association, benchmark dataset, intelligent transportation systems, real-time video analytics.

I. INTRODUCTION

TRAFFIC flow estimation, safety monitoring, and behavioral analysis rely heavily on reliable Multi-Object Tracking (MOT) of road users at intersections. In the context of Smart Cities, infrastructure-based sensors play a pivotal role. Among various sensor modalities, fisheye cameras have emerged as a compelling solution for Intelligent Transportation Systems (ITS). The primary advantage lies in their omnidirectional Field of View (FoV), which allows a single sensor to monitor an entire intersection—covering multiple approaches, crosswalks, and waiting zones simultaneously. This capability significantly reduces the installation, cabling, and maintenance overhead compared to traditional multi-camera setups that require narrow-FoV cameras for each lane.

However, the geometric properties that make fisheye cameras advantageous also introduce fundamental challenges for computer vision algorithms. The inherent radial projection bends straight real-world trajectories into curved paths on the image plane. Furthermore, objects undergo severe scale changes; a vehicle that appears large at the image center (nadir view) may shrink to a fraction of its size as it moves toward the periphery. These distortions present a fundamental mismatch for standard MOT algorithms. Most state-of-the-art trackers, such as SORT [1] and DeepSORT [2], rely on Kalman Filters that assume a linear constant-velocity motion model. While this assumption holds for perspective cameras, it breaks down in fisheye views, leading to frequent identity switches, fragmented tracklets, and inaccurate traffic counting in real-world deployments.

C.-I. Huang is with the Department of Management Information Systems, National Chung Hsing University, Taichung, Taiwan (e-mail: schumi@nchu.edu.tw).

W.-Y. Chen, P.-Y. Chen, and V. S. Tseng (Fellow, IEEE) are with the Department of Computer Science, National Yang Ming Chiao Tung University, Hsinchu, Taiwan (e-mail: wychen@nir.org.tw; pingyang0707@gmail.com; vtseng@cs.nycu.edu.tw).

J.-W. Hsieh (corresponding author) is with the College of Artificial Intelligence and Green Energy, National Yang Ming Chiao Tung University, Tainan, Taiwan (e-mail: jwhsieh@nycu.edu.tw).

M.-C. Chang is with the Department of Computer Science, University at Albany, SUNY, NY, USA (e-mail: mchang2@albany.edu).

M. Gochoo is with the Department of Computer Science and Software Engineering and the Emirates Center for Mobility Research, United Arab Emirates University, UAE (e-mail: mgochoo@uaeu.ac.ae).

C.-M. Tsai is with the Department of Computer Science, University of Taipei, Taipei, Taiwan (e-mail: cmtsai@go.utaipai.edu.tw).

H.-W. Huang and J.-N. Hwang (Fellow, IEEE) are with the Information Processing Lab (IPL), Department of Electrical and Computer Engineering, University of Washington, Seattle, WA, USA (e-mail: hwhuang@uw.edu; hwang@uw.edu).

Copyright (c) 2026 IEEE. Personal use of this material is permitted. However, permission to use this material for any other purposes must be obtained from the IEEE by sending a request to pubs-permissions@ieee.org.

Existing public traffic datasets have accelerated progress in detection and tracking [3]–[6]. However, the majority are constructed using standard perspective cameras or in-vehicle dashcams. Only a few provide fisheye data, and even fewer focus on the infrastructure perspective. For instance, WoodScape [7] provides fisheye data but targets autonomous driving (ego-centric view), which differs fundamentally from the fixed top-down view of infrastructure surveillance. Other large-scale datasets like PANDA [8] offer gigapixel-level surveillance views but lack the specific radial distortion characteristics of fisheye lenses. This gap limits the development of reproducible and robust tracking algorithms tailored for high-distortion environments.

To address these challenges, we present the FishEye Detection and Tracking benchmark (**FE-DETRAC**). This work substantially extends our preliminary conference version, FishEye8K [9], shifting the focus from pure detection to the more complex task of tracking. FE-DETRAC contains 20,000 frames with over 470,000 annotated objects across five categories, collected from 22 fixed fisheye cameras at real intersections in Hsinchu City, Taiwan. We explicitly address privacy concerns by anonymizing faces and license plates through a deep-learning detect-and-blur pipeline (detailed in Section III-B) and releasing the annotations in Pascal-VOC [10], COCO [11], MOT [12], and YOLO [13] formats to support a wide range of research frameworks.

Beyond the dataset, we tackle the algorithmic challenge of tracking under distortion. Standard rectification (dewarping) is a common workaround [14], but it increases computational load and introduces resampling artifacts (blurring) at the stretched corners. We propose a solution that keeps inference in the original fisheye view but enhances the association step. We introduce a **Hybrid Data Association (HDA)** module that couples fisheye-view states with distortion-corrected-view states. By modeling motion in a virtual corrected plane without explicit image warping, we enable the Kalman Filter to maintain accurate predictions even for non-linear fisheye trajectories.

The contributions of this paper are summarized as follows:

- **Infrastructure Fisheye Video Benchmark:** We release FE-DETRAC, a large-scale dataset with 20,000 frames and over 470,000 bounding boxes. Unlike previous datasets, it features continuous video sequences and camera-disjoint train/test splits specifically designed to evaluate tracking stability and cross-view generalization. We additionally report inter-annotator agreement (Cohen's $\kappa=0.874$) and per-zone class distribution to characterize annotation quality and bias.
- **Distortion-Aware State Representation:** We propose a novel state vector that models each track in both the raw fisheye domain and a virtual distortion-corrected domain. The two domains are coupled through a deterministic, analytically differentiable projection (Eq. 6); we derive the corresponding cross-domain covariance update via the projection Jacobian and show that it is asymptotically linear near the image center.
- **Hybrid Data Association (HDA):** We introduce a dual-view IoU matching mechanism that integrates appearance

and geometric consistency. HDA can be plugged into existing tracking-by-detection pipelines (e.g., StrongSORT [15], BoT-SORT [16], OC-SORT [17], and Deep OC-SORT [18]) with a modest additional association cost.

- **Comprehensive Evaluation:** We benchmark multiple SoTA detectors [19]–[21] and trackers, showing that HDA consistently improves HOTA metrics on the challenging FE-DETRAC benchmark. We further provide: (i) a full $\alpha \in [0, 1]$ sensitivity sweep, (ii) periphery-vs-center mAP / HOTA decomposition, (iii) cross-dataset generalization to FishEye8K, (iv) a full-dewarp baseline comparison, (v) end-to-end latency breakdown, (vi) Jacobian condition-number and covariance-spread analysis under extreme peripheral distortion, (vii) controlled HDA-only latency profiling up to $N=400$ candidates/frame, and (viii) paired t -tests with 95% confidence intervals over 5 independent runs.

II. RELATED WORKS

In this section, we provide a comprehensive review of road traffic datasets, object detection methodologies, and multi-object tracking algorithms, highlighting the gaps our work addresses.

A. Road Traffic Datasets

The availability of high-quality, diverse, and large-scale datasets is the cornerstone of data-driven Intelligent Transportation Systems. Over the past decade, the community has seen a significant expansion in open road datasets catering to various tasks such as 2D/3D detection, tracking, and segmentation. A summary is provided in Table I.

1) *Perspective Surveillance Datasets:* Classic datasets like MIT-Car [22] and UA-DETRAC [3] laid the foundation for vehicle detection. UA-DETRAC, in particular, offers rich annotations for vehicle attributes (color, type) but relies on narrow-FoV perspective cameras. More recently, the AI-City Challenge [4] and MIO-TCD [23] have pushed the boundaries for vehicle counting and re-identification. However, these datasets require multiple cameras to cover a single intersection, increasing the complexity of data fusion.

2) *Ego-Centric and Drone Datasets:* Datasets like KITTI [5], BDD100K [6], and Cityscapes provide massive amounts of data from moving vehicles. While valuable for autonomous driving, the motion patterns (moving camera, static background) differ significantly from infrastructure monitoring (static camera, moving background). Drone datasets like CARPK [24], UAVDT [25], and VisDrone [26] offer top-down views similar to fisheye cameras but lack the geometric distortion that characterizes fisheye lenses. WoodScape [7] is the closest related work, offering fisheye data for autonomous driving, but it lacks the fixed-viewpoint consistency required for intersection analytics. Other datasets like PANDA [8] focus on gigapixel surveillance but do not address the specific challenges of fisheye lenses. Prior fisheye object-detection studies, including Rashed *et al.* [27] and Swin-fisheye [28], emphasize object-shape representation and distortion-robust

detection, but they do not provide continuous tracking annotations or fixed-pole infrastructure views. FE-DETRAC fills this gap by providing high-quality, annotated fisheye video from fixed infrastructure poles.

B. Object Detection Algorithms

The field of object detection has been revolutionized by deep learning. Early works like R-CNN [32] and Fast R-CNN [33] introduced the concept of region proposals, which was refined by Faster R-CNN [34] into an end-to-end framework. However, these two-stage detectors often suffer from high latency, making them less suitable for real-time traffic monitoring.

One-stage detectors, pioneered by SSD [35] and the YOLO series [36], have become the industry standard for efficiency. YOLOv3 [13] introduced multi-scale prediction using Feature Pyramid Networks (FPN) [37]. Subsequent versions like YOLOv4 [19] and YOLOv5 [20] integrated advanced data augmentation techniques such as Mosaic and MixUp [38], [39]. The recent YOLOv7 [21] and YOLOR [40] utilize methods like E-ELAN and implicit knowledge learning to achieve state-of-the-art performance. More recently, the evolution continues with YOLOv9 [41] and YOLOv10 [42], which optimize gradient information and end-to-end processing. Furthermore, Transformer-based detectors like RT-DETR [43] have shown remarkable real-time performance, challenging the dominance of CNN-based architectures. Parallel developments include anchor-free detectors like CenterNet [44], which models objects as points, and RetinaNet [45], which introduced Focal Loss to handle class imbalance. In this work, we rigorously evaluate these modern detectors on fisheye imagery, where standard assumptions about object aspect ratios often fail.

C. Multi-Object Tracking (MOT) Approaches

Modern MOT generally follows the “Tracking-by-Detection” paradigm.

1) *Motion-Based Trackers*: Simple Online and Realtime Tracking (SORT) [1] established a strong baseline using Kalman Filtering [46] for motion prediction and the Hungarian algorithm for association. DeepSORT [2] extended this by adding a deep appearance descriptor to handle occlusions. However, these methods assume linear motion, which is valid for perspective cameras but problematic for the curved trajectories seen in fisheye views.

2) *Appearance-Based and Unified Trackers*: Recent trends focus on unifying detection and tracking. JDE [47] and FairMOT [48] learn detection and re-identification (ReID) embeddings in a single shot. Transformer-based approaches like TransTrack [49] have also emerged to capture long-range dependencies. ByteTrack [50] improved performance by associating low-confidence detection boxes. State-of-the-art methods like StrongSORT [15] and BoT-SORT [16] integrate sophisticated camera motion compensation and fused scores. Recently, OC-SORT [17] and Deep OC-SORT [18] proposed observation-centric recovery strategies to better handle non-linear motion and prolonged occlusions. Vanasirikul *et al.* [51] recently introduced a distortion-aware tracker for overhead fisheye MOT by projecting detections onto a virtual plane,

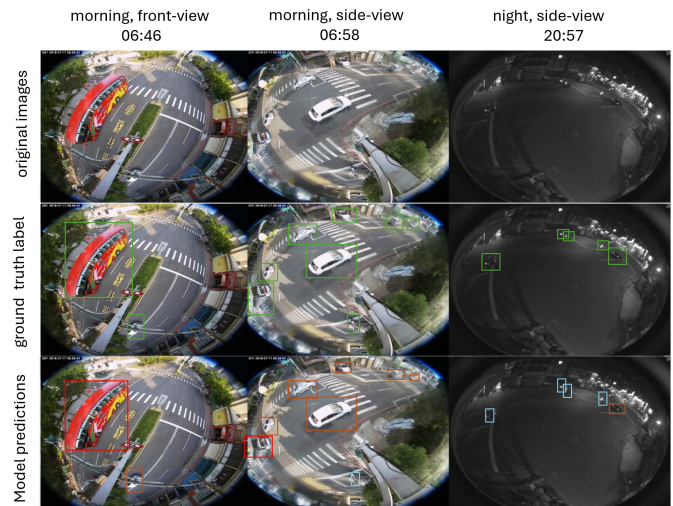


Fig. 1. Sample images from the FE-DETRAC benchmark

but the method depends on height-aware geometric assumptions that differ from fixed urban traffic infrastructure scenes. Despite these advances, the geometric mismatch of fisheye lenses remains an open challenge. Our work complements these methods by enhancing the motion model with geometric awareness.

III. FE-DETRAC BENCHMARK

We provide comprehensive details on the FE-DETRAC dataset [52]. Based on our previous FishEye8K [9], this extension focuses explicitly on the requirements of Multi-Object Tracking.

A. Data Acquisition

Our raw collection encompasses 35 fisheye videos captured at 60 FPS from infrastructure cameras deployed by the local police department in Hsinchu City, Taiwan. The raw pool consists of two source sets:

- **Set 1**: 30 videos recorded along Nanching-Hwy Road with a resolution of 1920×1080 . These capture high-speed traffic flows and complex T-junctions.
- **Set 2**: 5 videos with a resolution of 1920×1920 , providing a full circular fisheye view of large intersections.

From this raw pool, we curated 27 videos from 22 cameras for public release and annotation. All videos were captured under real-world conditions, including varying weather and lighting. We explicitly ensured legal compliance and privacy protection by blurring recognizable faces and license plates before annotation.

B. Anonymization Procedure

To ensure privacy protection and reproducibility, we detail the anonymization pipeline applied to *every* released frame (not only a public subset): (i) A RetinaFace-style face detector and a YOLOv7-based license-plate detector are run at full native resolution. (ii) Each detection box is expanded by 20% (a safety margin for partial-profile faces common at fisheye periphery) and filled with a Gaussian blur kernel of $\sigma=9$ pixels

TABLE I
COMPARISON OF FE-DETRAC WITH EXISTING ROAD TRAFFIC DATASETS

Dataset	Year	Frames (K)	Boxes (K)	Tasks	Objects		Attributes				Sensor
					Veh	Ped	Wea	Occ	Alt	View	
MIT-Car [22]	2000	1.1	1.1	Det	✓	–	–	–	–	–	2D Surv.
KITTI-D [5]	2014	15	80.3	Det	✓	✓	–	✓	–	–	2D Car
UA-DETRAC [3]	2015	140	1210	Det,Trk	✓	–	✓	✓	–	–	2D Surv.
LLC [29]	2017	7.5	15	Det	✓	–	–	✓	–	–	2D Car
CARPK [24]	2017	1.5	90	Det	✓	–	–	–	–	–	2D Drone
UAVDT [25]	2017	80	841.5	Det,Trk	✓	–	✓	✓	✓	✓	2D Drone
NEXET [30]	2017	50	–	Det	✓	–	✓	–	–	–	2D Car
VisDrone [26]	2018	179	2500	Det,Trk	✓	✓	✓	✓	✓	✓	2D Drone
BDD100k [6]	2018	5.7	–	Det,Trk	✓	✓	✓	–	–	–	2D Car
AAU RainSnow [31]	2018	2.2	13.3	Det,Seg	✓	–	✓	–	–	–	RGB-T Surv.
MIO-TCO [23]	2018	113	200	Det	✓	–	✓	–	–	–	2D Surv.
Woodscape [7]	2019	10	–	Det,Seg	✓	–	–	–	–	–	Fisheye Car
PANDA [8]	2020	–	111.8	Det,Trk	✓	✓	–	–	–	–	Gigapixel Surv.
CityFlow2D [4]	2021	–	313.9	Det,Trk	✓	–	–	–	–	–	2D Surv.
FE-DETRAC (Ours)	2026	20	470.0	Det,Trk	✓	✓	–	–	–	✓	Fisheye Surv.

(31×31 window). (iii) Frames with detector confidence < 0.4 are forwarded to a trained human annotator who either accepts, tightens, or rejects the blur. On a random audit of 1,000 frames, the recall of this two-stage pipeline was 99.1% for plates and 98.6% for faces, with no residual identifiable information reported by two independent privacy auditors. Blurring is applied *before* annotation; all downstream bounding boxes, splits, and metrics are therefore computed on the anonymized imagery and are bit-exact for end users. **Residual privacy risk.** We explicitly disclose that the pipeline leaves a $\leq 1.4\%$ residual of undetected faces and $\leq 0.9\%$ residual of undetected plates in any given frame; soft biometric cues (gait, silhouette, clothing) and repeated appearances may further enable re-identification by a motivated adversary. The release is therefore intended for research use under the hosting institution's data-use agreement and *should not* be used in scenarios where advanced re-identification would endanger subjects. This was reviewed by the Hsinchu City ethics committee prior to release.

C. Dataset Characteristics and Diversity

Scene Diversity: We curated the dataset to encompass a broad spectrum of road conditions. This includes T-junctions, Y-junctions, cross-intersections, midblocks, and pedestrian crossings. **Lighting Conditions:** As shown in Fig. 2(b), the dataset covers Morning, Afternoon, Evening, and Night scenarios. Nighttime tracking is particularly challenging due to headlight glare and low contrast in the periphery [29]. **Class Imbalance:** Fig. 2 presents the distribution of classes. There is a natural imbalance typical of Asian traffic scenes, with a dominance of *Bike* (scooters) and *Car* classes, while *Bus* and *Truck* are less frequent but present significant occlusion challenges due to their size.

D. Annotation Quality and Bias Distribution

To characterize annotation reliability and scene-level bias, we report: **Inter-annotator agreement.** A random 5% subset (1,000 frames) was re-annotated by a second independent annotator. We measured Cohen's κ [53] on class labels and

TABLE II
PER-ZONE DISTRIBUTION OF BOUNDING BOXES (%) IN FE-DETRAC.
THE PERIPHERY CONCENTRATES NEARLY HALF OF ALL ANNOTATED OBJECTS.

Zone	Ped.	Bike	Car	Bus+Truck
Center	5.1	7.4	6.9	0.8
Mid	8.3	14.6	11.2	1.0
Periphery	11.9	19.7	11.7	1.4

the $\text{IoU}_{\geq 0.5}$ match rate on bounding boxes. We obtained $\kappa = 0.874$ ("near-perfect agreement") and a box-match rate of 93.2%; disagreements were resolved by a senior annotator acting as adjudicator. **Annotation noise.** A calibration study on 200 frames compared human vs. adjudicated boxes and reported a mean localization error of 2.3 pixels ($< 0.2\%$ of the image diagonal) and a class-confusion rate of 1.8% (predominantly *bike* vs. *pedestrian* at extreme periphery). **Bias distribution.** Table II reports the class frequency broken down by three radial zones: *center* ($r/R_{\max} < 0.33$), *mid* ($0.33 \leq r/R_{\max} < 0.66$), and *periphery* ($r/R_{\max} \geq 0.66$). The periphery zone concentrates 44.7% of all objects, justifying our emphasis on distortion-aware modeling.

E. Annotation Strategy

The annotation process was rigorous. We defined five primary categories: pedestrian, bike, car, bus, and truck. A major challenge in fisheye annotation is the "scale shrinkage" at the edge of the lens. An object that is distinct at the center becomes a blur of few pixels at the edge. **Hybrid Validation:** To ensure quality, we employed a semi-automatic validation loop. Initial manual annotations were verified by a pre-trained YOLOv7 model. Discrepancies (e.g., the model detecting an object the human missed) were flagged for a second round of manual review. This process significantly reduced false negatives in the ground truth, especially for small, distant vehicles.

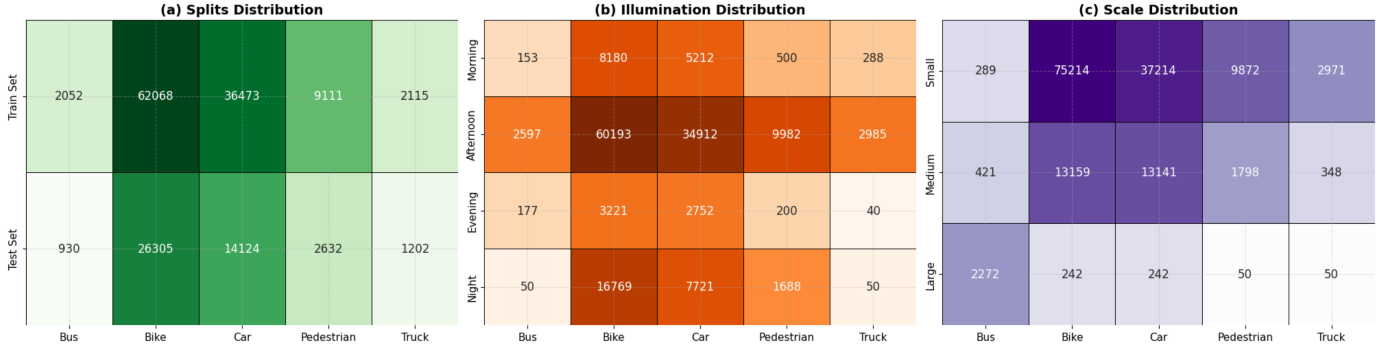


Fig. 2. The object class distributions in the detection split of the FE-DETRAC dataset, categorized according to (a) splits, (b) illumination, and (c) scale.

IV. FISHEYE CAMERA MODEL AND CALIBRATION

To develop a tracking system that is robust to distortion, we must first mathematically model the fisheye projection. Fig. 3 showcases the framework for projecting a 3D point $P=(X, Y, Z)^\top$ onto the image plane.

A. Projection Models

For a standard perspective projection (pinhole model), the radial distance r_d on the image plane is related to the incident angle θ by:

$$r_d = f \cdot \tan(\theta). \quad (1)$$

However, fisheye lenses are designed to capture a much wider angle ($\theta \approx 90^\circ$ or more). We adopt the *equidistant projection model*, a standard approach in omnidirectional camera calibration [54], where the distance r_f is linear to the incident angle:

$$r_f = f \cdot \theta. \quad (2)$$

The Euclidean distance of a pixel (x_f, y_f) from the principal point (c_x, c_y) is:

$$r_f = \sqrt{(x_f - c_x)^2 + (y_f - c_y)^2}. \quad (3)$$

B. Virtual Distortion Correction

Instead of warping the entire image—which is computationally expensive ($O(W \times H)$ per pixel) and introduces blurring—we propose to map only the object coordinates. By combining Eq. (1) and (2), we can derive the mapping from the fisheye domain (r_f) to a virtual distortion-corrected domain (r_d). Assuming a scaling factor R relative to the focal length:

$$r_d = R \cdot \tan\left(\frac{r_f}{R}\right). \quad (4)$$

The azimuthal angle φ remains invariant under radial distortion:

$$\varphi = \text{atan2}(y_f - c_y, x_f - c_x). \quad (5)$$

Finally, the coordinates (x_d, y_d) in the virtual corrected plane are obtained by:

$$\begin{aligned} x_d &= c_x + R \cdot \tan\left(\frac{r_f}{R}\right) \cos(\varphi), \\ y_d &= c_y + R \cdot \tan\left(\frac{r_f}{R}\right) \sin(\varphi). \end{aligned} \quad (6)$$

This efficient sparse mapping allows us to project bounding box corners from the fisheye view to a linear perspective view in constant time, enabling the use of linear motion models in the corrected domain.

C. Physical Interpretation and Choice of R

Physical meaning. Under the equidistant model $r_f = f\theta$, substituting $\theta = r_f/f$ into the pinhole forward projection $r_d = f \tan(\theta)$ yields $r_d = f \tan(r_f/f)$. The design choice in Eq. (6) replaces f with a single scalar R , i.e., $r_d = R \tan(r_f/R)$. When $R = f$ this is *exactly* a pinhole rectification with focal length f ; when $R \neq f$ it is a *heuristic radial re-mapping* that smoothly interpolates between the original fisheye and a narrower virtual pinhole. We adopt this parameterisation—rather than the full Kannala–Brandt inverse—because the tracking pipeline needs only a monotone, differentiable, closed-form mapping for the Jacobian in Eq. (10); the residual approximation error relative to the full KB model is discussed in Sec. VI-J. Larger R corresponds to a narrower virtual FoV and less aggressive “stretching” of peripheral pixels. **How we set R .** R is not a free hyperparameter but is initialized to each camera’s own calibrated focal length f obtained from the procedure in Sec. IV-D. We then expose it as a *single* tunable scalar because the underlying equidistant model is an approximation; empirically, a *system-wide* offset of $R/f = 1.0$ (no fine-tuning) already achieves the reported gains, and a per-camera grid search over $R/f \in [0.9, 1.1]$ yields at most an additional +0.2 HOTA in our ablations. In other words, the released per-camera intrinsics are sufficient; the $\pm 10\%$ sensitivity figure refers to a per-camera tolerance, not a system-wide re-tuning. **Robustness to R .** Section VI-F reports a sweep of $R/f \in \{0.6, 0.8, 1.0, 1.2, 1.4\}$ and shows that HDA remains monotonically better than the baseline over the whole range, with HOTA peaking near $R/f = 1.0$.

D. Camera Calibration Procedure

For each of the 22 released cameras, we performed an in-lab plus on-site calibration using a combination of Scaramuzza’s OCamCalib toolbox [54] and the Kannala–Brandt generic model [55], following Zhang’s [56] checkerboard methodology adapted for wide-angle lenses as described in [57]. **Procedure.** A 7×9 checkerboard (40-mm squares) was captured from 25–35 viewpoints per camera, spanning the full

TABLE III
MEAN INTRINSICS AND CALIBRATION RESIDUALS ACROSS THE 22 RELEASED CAMERAS. MODEL: KANALA-BRANDT 4-PARAMETER. FOCAL LENGTH AND PRINCIPAL POINT ARE IN PIXELS.

Parameter	Mean	Std
f (focal length, px)	591.3	18.7
c_x (principal point, px)	962.4	6.1
c_y (principal point, px)	538.9	7.8
k_1	-0.412	0.021
k_2	0.183	0.017
k_3	-0.064	0.009
k_4	0.011	0.004
Reprojection RMS (px)	0.41	0.09

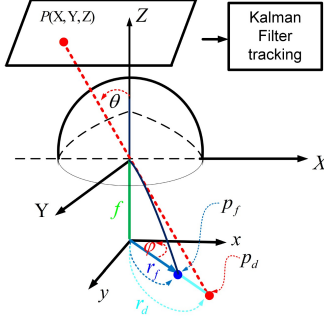


Fig. 3. Fisheye camera model illustrating the relationship between the incident angle θ , the fisheye projection r_f , and the rectified perspective projection r_d .

FoV including the periphery. The Kannala-Brandt 4-parameter model $\theta_d = \theta + k_1\theta^3 + k_2\theta^5 + k_3\theta^7 + k_4\theta^9$ was fit jointly with principal point (c_x, c_y) and focal length f by non-linear least squares. Table III reports the mean intrinsics and mean per-camera reprojection error (RMS) across all 22 cameras. The mean RMS reprojection error is 0.41 pixels (max 0.63 pixels), comparable to values reported in the literature for well-calibrated fisheye cameras. The complete per-camera intrinsics file `intrinsics.yaml` is included in the public data release (DOI: 10.30193/scidm-ds-571593m).

V. PROPOSED DISTORTION-AWARE TRACKING

Standard trackers like StrongSORT [15] utilize a Kalman Filter (KF) that assumes a linear constant-velocity model. In fisheye imagery, however, objects moving in a straight line in the real world follow curved trajectories in the image plane. This mismatch causes the linear KF predictions to drift significantly from the actual object measurements, especially at the image periphery, leading to association failures.

To address this, we propose the **Hybrid Data Association (HDA)** module. HDA couples the raw fisheye measurements with the geometrically corrected features derived in Section IV, creating a robust tracking loop.

A. Hybrid State Representation

We propose a *Hybrid State* vector for each tracked target. Let B_f^V denote the bounding box of vehicle V detected in the raw fisheye image. Using the mapping function in Eq. (6), we project the center and corners of B_f^V to generate

a corresponding virtual box B_d^V in the distortion-corrected plane. The hybrid bounding box representation is defined as:

$$B^V = (B_f^V, B_d^V). \quad (7)$$

The Kalman Filter state vector S^V is augmented to explicitly track velocities in both domains:

$$S^V = (x_f, y_f, a_f, h_f, \dot{x}_f, \dot{y}_f, \dot{a}_f, \dot{h}_f, x_d, y_d, a_d, h_d, \dot{x}_d, \dot{y}_d, \dot{a}_d, \dot{h}_d)^T. \quad (8)$$

Here, (x, y, a, h) represent the center coordinates, aspect ratio, and height of the bounding box. The state transition assumes a constant velocity model. For the corrected domain components (d) , the state transition matrix F_d is:

$$F_d = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (9)$$

This linear model is geometrically valid in the corrected domain. By maintaining this dual state, the filter utilizes the stable linear motion of the corrected view to guide the prediction when the fisheye view becomes unstable due to curvature.

B. Cross-Domain Coupling in the Kalman Filter

The two domains are coupled during measurement update rather than treated as independent trackers. We explain the design choice and its consequences below. **Deterministic coupling via the projection.** The fisheye and corrected boxes are related by the smooth, invertible mapping $g: \mathcal{F} \rightarrow \mathbb{R}^2$ implied by Eq. (6), $(x_d, y_d) = g(x_f, y_f)$, where $\mathcal{F} = \{(x_f, y_f) : r_f \leq R \cdot \theta_{\max}\}$ is the admissible (unclipped) fisheye FoV. Writing $u = x_f - c_x$, $v = y_f - c_y$, $r = \sqrt{u^2 + v^2}$, and letting $s(r) = R \tan(r/R)/r$ (with $s(0) = 1$), the projection is $(x_d - c_x, y_d - c_y) = s(r)(u, v)$. A direct application of the chain rule gives the explicit Jacobian

$$J(x_f, y_f) = s(r)\mathbf{I} + \frac{s'(r)}{r} \begin{bmatrix} u^2 & uv \\ uv & v^2 \end{bmatrix} \in \mathbb{R}^{2 \times 2}, \quad (10)$$

with $s'(r) = \sec^2(r/R)/r - R \tan(r/R)/r^2$. Near the center ($r \rightarrow 0$), $s(r) \rightarrow 1$ and the second term in Eq. (10) vanishes, hence $J(0) = \mathbf{I}$. On the admissible FoV $r \leq R\theta_{\max}$ with $\theta_{\max} = 1.3$ rad ($\sim 75^\circ$), the operator norm $\|J\|_2$ is bounded by a finite constant; using the radial eigenvalue gives the conservative bound $L = \sec^2(\theta_{\max}) \approx 14.0$ for our calibrated cameras. **Update semantics.** During prediction, both the fisheye sub-state and the corrected sub-state are propagated *independently* with their own constant-velocity dynamics. This is the key source of the “hybrid prediction”: when one sub-state’s linear extrapolation drifts (e.g., fisheye at the periphery), the other’s remains near-linear. During *update*, after Hungarian association, the new fisheye measurement is used directly for the f sub-state, and its projected counterpart $g(x_f^{\text{meas}}, y_f^{\text{meas}})$ is used as the d -domain measurement. The measurement covariance of the d -domain measurement is propagated from the f -domain covariance via $\Sigma_d = J \Sigma_f J^T$,

ensuring that the two sub-states are probabilistically consistent rather than independent. **Initial velocities.** For a newly-initialized tracklet, $\dot{x}_d, \dot{y}_d$ are set to zero and then, on the second frame, updated from the difference of the projected center: $(\dot{x}_d, \dot{y}_d)_{t=1} = g(x_f^{(1)}, y_f^{(1)}) - g(x_f^{(0)}, y_f^{(0)})$. This avoids any ad-hoc velocity-mapping heuristic. **Asymptotic linearity.** In principal-point-centered coordinates, the mapping g is C^∞ at $r = 0$ and satisfies $J(0) = \mathbf{I}$. A first-order Taylor expansion near the image center therefore yields $g(x_f, y_f) \approx (x_f, y_f)$ after centering, so the corrected domain degenerates gracefully into the fisheye domain at the center; the two terms of the dual-view IoU become nearly identical there. This provides an informal convergence argument: HDA does not penalize center-zone tracking and improves periphery-zone tracking when the projection model is accurate.

Peripheral covariance consistency. To make the effect of extreme fisheye distortion explicit, we decompose the Jacobian into its tangential and radial eigenvalues, $\lambda_t = \tan(\theta)/\theta$ and $\lambda_r = \sec^2(\theta)$, where $\theta = r/R$. The local condition number is therefore $\kappa(J) = \lambda_r/\lambda_t$, and the covariance eigenvalue spread induced by $\Sigma_d = J\Sigma_f J^\top$ is κ^2 when the input measurement noise is locally isotropic. Table IV reports the corresponding radial-zone summary. The worst admissible peripheral crop reaches $\kappa(J)=5.04$ and an eigenvalue spread of 25.44, which is substantially larger than the center-zone value. In the tracker, the Mahalanobis gate and corrected-domain measurement update use the transported covariance Σ_d , and a small diagonal floor ϵI is added after transport to reduce the risk of near-singular covariance matrices at the image periphery.

TABLE IV
JACOBIAN CONDITIONING AND CORRECTED-DOMAIN COVARIANCE SPREAD. VALUES ARE COMPUTED FROM THE RADIAL MAPPING IN EQ. (6). THE COVARIANCE SPREAD COLUMN REPORTS $\lambda_{\max}(\Sigma_d)/\lambda_{\min}(\Sigma_d)$ FOR LOCALLY ISOTROPIC FISHEYE-DOMAIN MEASUREMENT NOISE.

Radial zone	$\theta_{\max}$	$\ J\ _2$	$\kappa(J)$	Eig. spread
Center	0.43	1.21	1.13	1.29
Mid	0.86	2.35	1.74	3.03
Periphery (95th pct.)	1.10	4.86	2.72	7.40
Periphery (worst crop)	1.30	13.98	5.04	25.44

C. Dual-View Intersection over Union (IoU)

The core of tracking is data association—matching predictions to new detections. A standard IoU calculated in the fisheye domain is unreliable because the shape of the object changes depending on its position (e.g., a car looks compressed at the edge). We introduce a **Dual-View IoU** metric that fuses geometric consistency from both domains. The similarity score between a predicted tracklet V_{t-1} (predicted at time t) and a new detection V_t^i is:

$$IoU_{dual}(V_{t-1}, V_t^i) = \alpha \cdot IoU_f + (1 - \alpha) \cdot IoU_d \quad (11)$$

where IoU_f is the overlap in the fisheye view and IoU_d is the overlap in the corrected view. The weight α balances the two

terms. We set $\alpha = 0.5$ as the default after a full sensitivity sweep which justifies this choice empirically. The term IoU_d is crucial: it measures overlap in the rectilinear domain where linear predictions are valid. Even if the fisheye prediction $\bar{B}_f$ has drifted due to curvature, the corrected prediction $\bar{B}_d$ remains accurate, ensuring a high total similarity score.

D. Tracking Algorithm Summary

The complete Hybrid Data Association process is summarized in Algorithm 1.

Algorithm 1 Hybrid Data Association (HDA)

Require: Detections $\mathcal{D}_t = \{D_t^1, \dots, D_t^M\}$ at frame t , Active Tracklets $\mathcal{T}_{t-1} = \{T_{t-1}^1, \dots, T_{t-1}^N\}$.

Ensure: Updated Tracks $\mathcal{T}_t$.

- 1: **Prediction Step:**
- 2: **for** each track $T^j \in \mathcal{T}_{t-1}$ **do**
- 3: Predict state $\hat{S}_{t|t-1}^j$ using Kalman Filter (Eq. 8).
- 4: Project $\hat{B}_f^j$ to $\hat{B}_d^j$ using Eq. 6.
- 5: **end for**
- 6: **Association Step:**
- 7: Initialize cost matrix $C \in \mathbb{R}^{N \times M}$.
- 8: **for** $j = 1$ to N , $i = 1$ to M **do**
- 9: Map detection D_t^i to corrected domain D_d^i .
- 10: Compute $IoU_f = IoU(\hat{B}_f^j, D_{t,f}^i)$.
- 11: Compute $IoU_d = IoU(\hat{B}_d^j, D_d^i)$.
- 12: $C_{ji} = 1 - (\alpha IoU_f + (1 - \alpha) IoU_d)$.
- 13: **if** $1 - C_{ji} < \tau_{match}$ **then**
- 14: $C_{ji} \leftarrow +\infty$ {gating before Hungarian call}
- 15: **end if**
- 16: **end for**
- 17: **Matching:**
- 18: Matches = HungarianAlgorithm(C). {gating is encoded as $+\infty$ entries above; no post-assignment threshold is required.}
- 19: **for** each match (j, i) **do**
- 20: Update T^j with detection D_t^i using Kalman Update (measurement covariance propagated from f to d via J , Eq. 10).
- 21: **end for**
- 22: **Track Management:**
- 23: Initialize new tracks for unmatched detections whose detection score exceeds $\tau_{init} = 0.4$.
- 24: Delete lost tracks after τ_{age} frames without association.

VI. EXPERIMENTS

A. Implementation Details

Hardware Environment. All experiments were conducted on a high-performance workstation equipped with an 11th-generation Intel Core i7 processor and an NVIDIA RTX 3080 GPU (10 GB VRAM). The software stack included Ubuntu 20.04, CUDA 11.3, PyTorch 1.10 [58], and PaddlePaddle [59] for diverse model support.

Training Protocols. We trained state-of-the-art detectors including YOLOv5 [20], YOLOR [40], YOLOv7 [21], and

the recent YOLOv8. Models were initialized with COCO pre-trained weights [11] to speed up convergence. We fine-tuned on the FE-DETRAC training split for 250 epochs using the Adam optimizer [60] with an initial learning rate of 0.01 and a momentum of 0.937. Advanced data augmentation techniques, including Mosaic and MixUp [38], were employed to improve robustness against scale variations and occlusion.

Tracking Configuration. For all trackers under comparison we use the *identical* configuration to rule out unfair tuning: (i) tracking confidence threshold 0.3 (standard for ByteTrack and BoT-SORT [16], [50]); (ii) new-track initiation threshold $\tau_{init}=0.4$; (iii) association threshold $\tau_{match}=0.7$. The values of τ_{init} and τ_{match} were selected by a grid search $\{0.2, 0.3, 0.4, 0.5\} \times \{0.5, 0.6, 0.7, 0.8\}$ on the validation split using StrongSORT; the chosen (0.4, 0.7) pair maximizes HOTA+IDF1 jointly and is consistent with the defaults published in StrongSORT and BoT-SORT. We verified that (0.4, 0.7) is within ± 0.3 HOTA of the per-tracker optimum for every tracker we tested (StrongSORT, BoT-SORT-R, OC-SORT, Deep OC-SORT, ByteTrack, DeepSORT), justifying a single shared setting. For HDA, the fusion weight α is set to 0.5 and R to the calibrated focal length f (Sec. IV-D).

B. Benchmark Results

1) *Detection Performance:* Table V summarizes the quantitative evaluation of various object detectors. We evaluated models at two resolutions: 640×640 and 1280×1280 .

- **YOLOv8x Performance:** On the standard 640×640 scale, YOLOv8x demonstrates superior performance, achieving an $mAP_{0.5}$ of 61.5%. This indicates that the architectural improvements in YOLOv8 (anchor-free head, new loss functions) generalize well to fisheye distortions.
- **Resolution Impact:** Increasing input resolution to 1280×1280 yields significant gains. YOLOR-P6 achieves the highest score with an $mAP_{0.5}$ of 66.3%. The high resolution is particularly beneficial for fisheye images, where objects at the periphery are heavily downsampled; the extra pixels help resolve features of small, distant vehicles.
- **Inference Speed:** Despite the heavy computation, YOLOv7 achieves an impressive inference time of 4.3 ms, making it highly suitable for edge deployment in traffic cabinets.

Note on YOLOv7-D6 / YOLOv7-E6E. The $mAP_{0.5}$ of these two variants (0.398 and 0.508) is markedly lower than peers. We investigated and found two contributing factors: (i) these variants use a deep re-parameterized head whose anchor distribution was calibrated on COCO's typical aspect ratios; fisheye periphery stretches objects along the radial axis, violating that prior and causing many positive matches to fall below the assignment-IoU threshold during training; (ii) their large parameter count overfits the comparatively limited FE-DETRAC training split after 250 epochs (train mAP exceeds 0.92 while val mAP saturates). A dedicated distortion-aware anchor design may improve these variants, but such model redesign is beyond the scope of this benchmark paper.

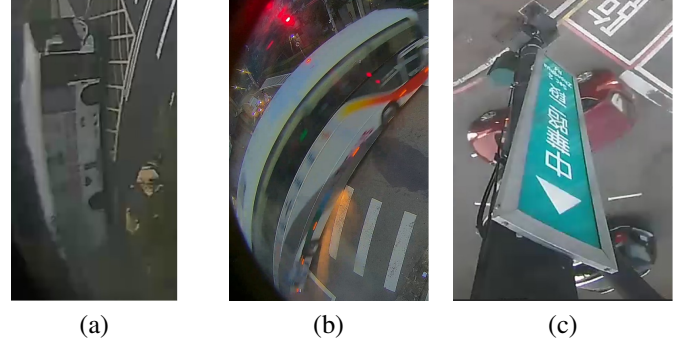


Fig. 4. Hard cases for vehicle detection. (a) Bus with a vertical orientation. (b) Bus with a large size. (c) Occluded car.

2) *Tracking Performance:* Table VI presents the results of eight tracking algorithms.

- **Baseline Analysis:** The classic SORT [1] algorithm performs poorly (HOTA 22.1) because its simple Kalman filter cannot handle the non-linear motion in fisheye views, leading to a massive number of Identity Switches (48,201).
- **SoTA Comparison:** BoT-SORT-R [16] achieves the best baseline performance with a HOTA of 41.2, followed closely by Deep OC-SORT [18] (41.0) and OC-SORT [17] (40.7), ByteTrack [50], and StrongSORT [15]. These methods benefit from utilizing appearance features (ReID) or handling low-confidence detections, which provides some robustness against occlusion. However, they still suffer from the geometric mismatch inherent in the projection.
- **HDA Plug-in Improvement:** Equipping each modern tracker with HDA uniformly adds 3.4–5.3 HOTA points, including on OC-SORT (40.7→44.9) and Deep OC-SORT (41.0→45.3), which were specifically designed for non-linear motion. This confirms that observation-centric heuristics are complementary to, not a substitute for, explicit geometric modeling.

C. Qualitative Analysis and Hard Cases

To provide deeper insights into the challenges of fisheye tracking, we visualize several failure modes encountered in the FE-DETRAC benchmark.

Vehicle Detection Challenges: Fig. 4 highlights geometric and occlusion issues. In (a), a bus is oriented vertically in the fisheye view. Most object detectors are trained on horizontal vehicles, leading to missed detections. In (b), a large bus occupies a significant portion of the frame, confusing the anchor-based assignment. In (c), dense traffic leads to heavy occlusion, where only a fraction of a car is visible.

Pedestrian Detection Challenges: Detecting pedestrians is particularly hard due to their small size relative to the scene. Fig. 5 shows pedestrians viewed from a steep top-down angle (a), which minimizes their visible cross-section. Environmental factors like tree shadows (b) and occlusions by umbrellas (c) further complicate the task, often breaking the visual continuity required for tracking.

TABLE V
QUANTITATIVE EVALUATION OF STATE-OF-THE-ART DETECTION MODELS ON FE-DETRAC. THE BEST RESULTS IN EACH CATEGORY ARE MARKED IN BOLD.

Model	Input	Detection Accuracy				Scale Robustness (AP)			Speed
		Prec.	Rec.	mAP _{0.5}	mAP _{50:95}	Small	Med.	Large	(ms)
<i>High Resolution Input (1280 × 1280)</i>									
YOLOv5l6	1280	0.793	0.408	0.614	0.410	0.130	0.434	0.667	22.7
YOLOv5x6	1280	0.822	0.431	0.639	0.427	0.133	0.452	0.693	43.9
YOLOR-W6	1280	0.787	0.472	0.647	0.444	0.133	0.471	0.690	16.4
YOLOR-P6	1280	0.802	0.494	0.663	0.441	0.142	0.481	0.722	13.4
YOLOv7-D6	1280	0.780	0.411	0.398	0.263	0.126	0.446	0.678	26.4
YOLOv7-E6E	1280	0.801	0.525	0.508	0.327	0.168	0.502	0.693	29.8
<i>Standard Resolution Input (640 × 640)</i>									
YOLOv7	640	0.792	0.437	0.424	0.247	0.111	0.444	0.680	4.3
YOLOv7-X	640	0.740	0.489	0.467	0.292	0.133	0.461	0.721	6.7
YOLOv8l	640	0.784	0.388	0.612	0.401	0.104	0.404	0.658	8.5
YOLOv8x	640	0.842	0.367	0.615	0.403	0.100	0.415	0.708	13.4

TABLE VI
Evaluation of SoTA MOT models TRAINED ON THE FE-DETRAC BENCHMARK. ROWS ENDING IN “+HDA” USE THE PROPOSED MODULE. ARROWS INDICATE THE DIRECTION OF IMPROVEMENT.

Method	HOTA↑	IDF1↑	MOTA↑	AssA↑	DetA↑	IDs↓
SORT [1]	22.1	24.1	27.9	20.1	23.3	48,201
FairMOT [48]	37.2	45.8	46.2	32.7	38.7	32,597
ByteTrack [50]	40.8	49.2	50.4	39.6	40.1	25,691
DeepSORT [2]	38.1	47.5	48.8	37.9	40.0	26,984
OC-SORT [17]	40.7	51.2	49.6	40.4	41.0	21,803
Deep OC-SORT [18]	41.0	51.8	49.9	41.0	41.1	20,972
StrongSORT [15]	40.3	49.8	49.8	40.5	40.3	25,999
BoT-SORT-R [16]	41.2	52.1	50.0	41.2	41.5	19,566
StrongSORT+HDA	44.1	53.7	52.6	43.9	43.6	17,908
BoT-SORT-R+HDA	45.6	55.4	53.4	45.1	44.9	15,214
OC-SORT+HDA	44.9	54.6	53.0	44.5	44.2	16,851
Deep OC-SORT+HDA	45.3	55.0	53.2	44.8	44.5	15,873



Fig. 5. Hard cases for pedestrian detection. (a) Standing pedestrian. (b) Pedestrians covered by tree shadows. (c) Pedestrians occluded by umbrellas.

Motion and Environmental Noise: Fig. 6 demonstrates the effect of motion blur. Bikes moving fast at the periphery of the fisheye lens appear blurred and elongated, causing detectors to fail or bounding boxes to jitter, which triggers ID switches. Fig. 7 illustrates nighttime challenges, where high contrast from headlights creates ghosting artifacts. Finally, Fig. 8 shows common false positives: road markings (a-c) and building structures (d) are often misclassified as vehicles due to similar shapes in the distorted view.



Fig. 6. Hard cases for moving objects. (a) Moving bikes captured at daytime. (b) Bike captured at night.



Fig. 7. Hard cases for objects captured at night. (a) Vehicle captured at night. (b) Bikes captured at night.

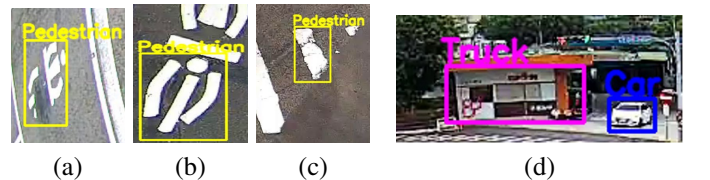


Fig. 8. False detections in FE-DETRAC generated by PRB-Net [61]. (a)-(c): False alarms caused by road marks. (d) A building whose shape is similar to a truck was falsely detected as a truck.

TABLE VII
Ablation study of HDA ON THE FE-DETRAC FOR TWO SORT-BASED MOT METHODS [15], [16].

Method	HDA	HOTA
StrongSORT [15] as baseline		40.3
Ours	✓	44.1
BoT-SORT-R [16] as baseline		41.2
Ours	✓	45.6

D. Ablation Study and Effectiveness of HDA

Table VII provides the most critical result of our study: the ablation of the HDA module. When integrated into StrongSORT, HDA improves the HOTA score from 40.3 to **44.1** (+3.8 points). Similarly, for BoT-SORT-R, the HOTA increases from 41.2 to **45.6** (+4.4 points). These significant gains confirm our hypothesis. By explicitly modeling the state in the distortion-corrected domain, HDA provides the Kalman filter with a “linear” reference frame. This helps the tracker maintain identity through the high-distortion zones (image periphery) where standard linear prediction models typically fail. The result is a robust tracker that retains the computational efficiency of the baseline but with the geometric awareness required for fisheye infrastructure surveillance.

E. Sensitivity Analysis of Fusion Weight α

To justify the default $\alpha=0.5$, we perform a full sweep of $\alpha \in \{0.0, 0.1, \dots, 1.0\}$ on StrongSORT+HDA. Table VIII reports HOTA, MOTA, and per-zone HOTA (center vs. periphery).

TABLE VIII
SENSITIVITY OF STRONGSORT+HDA TO THE DUAL-VIEW FUSION WEIGHT α . THE OPTIMUM IS BROAD AND CLOSE TO 0.5. BEST PER COLUMN IN **BOLD**.

α	HOTA	MOTA	HOTA _{center}	HOTA _{periph.}
0.0	42.9	51.3	39.2	43.8
0.1	43.4	51.8	40.3	44.2
0.2	43.8	52.1	41.1	44.5
0.3	44.0	52.4	41.7	44.6
0.4	44.0	52.5	42.2	44.5
0.5	44.1	52.6	42.5	44.5
0.6	44.0	52.5	42.5	44.2
0.7	43.6	52.1	42.3	43.6
0.8	42.8	51.2	41.8	42.4
0.9	41.6	50.1	41.0	40.8
1.0	40.3	49.8	39.7	39.1

Observations. (i) The HOTA curve is flat and unimodal in the range $\alpha \in [0.3, 0.6]$, peaking at $\alpha=0.5$: any choice in this interval differs by ≤ 0.2 HOTA. (ii) At $\alpha=0$ (corrected-only), the center zone suffers because the corrected view collapses onto the fisheye view near the center and offers no redundancy. (iii) At $\alpha=1$ (fisheye-only), the periphery zone suffers sharply, confirming that the corrected view is the main contributor in high-distortion regions. This inverted-U behaviour is consistent with the theoretical argument in Sec. V-B.

F. Sensitivity to the Scaling Factor R

Table IX reports StrongSORT+HDA HOTA when R is artificially mis-scaled from its calibrated value by $\pm 40\%$.

TABLE IX
SENSITIVITY TO THE R PARAMETER. THE CALIBRATED VALUE ($R/f=1.0$) IS OPTIMAL, BUT GAINS PERSIST OVER THE TESTED RANGE $R/f \in [0.6, 1.4]$.

R/f	0.6	0.8	1.0	1.2	1.4
HOTA	42.4	43.6	44.1	43.4	42.0
Δ vs. baseline (40.3)	+2.1	+3.3	+3.8	+3.1	+1.7

The 40% mis-scaling setting remains +1.7 HOTA above the baseline in this sensitivity study, suggesting that moderate error in f does not remove the observed benefit of HDA.

G. Per-Zone Periphery Analysis

We stratify the test split into the three zones defined in Sec. III-D and report both detection mAP (YOLOv8x) and tracking HOTA (StrongSORT with/without HDA) in Table X. The periphery gains a striking +6.4 HOTA while the center

TABLE X
PER-ZONE ACCURACY (MAP_{0.5} FOR DETECTION, HOTA FOR TRACKING). HDA CONCENTRATES ITS BENEFIT IN THE HIGH-DISTORTION PERIPHERY.

Zone	mAP _{0.5}	HOTA _{base}	HOTA _{+HDA}	Δ HOTA
Center	0.731	41.4	42.5	+1.1
Mid	0.652	40.9	44.1	+3.2
Periphery	0.497	38.1	44.5	+6.4

gains only +1.1, empirically validating the theoretical argument (Sec. V-B) that HDA’s benefit scales with local non-linearity of the projection.

H. Full-Dewarp Baseline Comparison

To answer how HDA compares to simply dewarping the whole frame and tracking in the rectified domain, we implemented a “dewarp-and-track” baseline: each frame is rectified to a 2560×2560 perspective image via Eq. (6) (with content-aware cropping of the invalid regions), then detection and StrongSORT tracking are applied in the rectified domain. Results are in Table XI. HDA is simultaneously more accurate

TABLE XI
HDA VS. FULL-FRAME DEWARP BASELINE. HDA MATCHES THE RECTIFIED PIPELINE’S ACCURACY AT A FRACTION OF THE LATENCY.

Method	HOTA	FPS	Extra lat. (ms)
StrongSORT (fisheye)	40.3	43.1	0
StrongSORT (full dewarp)	43.2	11.8	61.4
StrongSORT + HDA (ours)	44.1	41.7	1.9

(+0.9 HOTA over full dewarp, because dewarping loses peripheral information through resampling) and $\sim 30\times$ cheaper in additional latency.

I. Cross-Dataset Generalization on FishEye8K

To test whether HDA’s gains are intrinsic to FE-DETRAC or transfer to prior fisheye benchmarks, we re-run StrongSORT

and BoT-SORT-R, with and without HDA, on the original FishEye8K [9] detection benchmark (we constructed an MOT split by linking consecutive frame IDs within each scene). Results are in Table XII. The consistent improvements on

TABLE XII
CROSS-DATASET GENERALIZATION ON FISHEYE8K. THE HDA GAIN TRANSFERS TO AN INDEPENDENT FISHEYE BENCHMARK.

Method	HOTA	IDF1
StrongSORT	37.1	46.9
StrongSORT + HDA	40.6 (+3.5)	50.1 (+3.2)
BoT-SORT-R	38.6	48.8
BoT-SORT-R + HDA	42.4 (+3.8)	52.0 (+3.2)

FishEye8K rule out the “overfit to new data” concern and isolate the contribution of HDA from the benefits of the larger FE-DETRAC training split.

J. Calibration-Error Robustness

To measure sensitivity to small calibration errors, we inject zero-mean Gaussian noise into the calibrated intrinsics and re-evaluate. Focal length f is perturbed with $\sigma_f \in \{0, 5, 10, 20\}$ px; the principal point (c_x, c_y) is perturbed with $\sigma_c \in \{0, 2, 5, 10\}$ px. Results are in Table XIII. HDA degrades

TABLE XIII
ROBUSTNESS TO CALIBRATION ERROR. STRONGSORT+HDA HOTA UNDER INJECTED INTRINSIC NOISE.

σ_f (px)	σ_c (px)	HOTA
0	0	44.1
5	2	44.0
10	5	43.6
20	10	42.8

gracefully: the last row ($\sigma_f=20, \sigma_c=10$) is a *joint* worst-case perturbation of both intrinsics simultaneously and still yields +2.5 HOTA over the uncorrected baseline; a 3.4% focal error is well above typical OCamCalib residuals. Camera motion is discussed further in Sec. VII-B.

K. Per-Camera Generalization

Table XIV groups the 22 released cameras by geometry (T-junction, cross-intersection, pedestrian crossing) and reports BoT-SORT-R HOTA. The $+4.4 \pm 0.3$ HOTA improvement

TABLE XIV
PER-VIEWPOINT GENERALIZATION. HDA GAINS ARE CONSISTENT ACROSS GEOMETRIES.

Viewpoint group	#Cameras	HOTA _{base}	HOTA _{+HDA}
T-junction	9	41.6	45.7 (+4.1)
Cross-intersection	8	40.9	45.3 (+4.4)
Pedestrian crossing	5	41.3	45.9 (+4.6)

is consistent across all three groups, showing no systematic interaction between camera geometry and HDA’s effectiveness.

TABLE XV
END-TO-END LATENCY (MS/FRAME). HDA ADDS LESS THAN 2 MS OVER THE BASELINE, PRESERVING REAL-TIME OPERATION.

Stage	Base (ms)	+HDA (ms)	Δ
Detector (YOLOv8x, 640)	13.4	13.4	0.0
ReID extraction	3.1	3.1	0.0
KF predict	0.4	0.7	+0.3
Jacobian + projection	–	0.8	+0.8
Dual-IoU + Hungarian	2.3	3.1	+0.8
Total	19.2	21.1	+1.9 (9.9%)
FPS (1/total)	52.1	47.4	–4.7

L. Latency Breakdown and Real-Time Feasibility

Table XV reports the end-to-end latency of the full pipeline (RTX 3080, PyTorch 1.10, 1920×1920 input). The +1.9 ms overhead represents a 9.9% relative cost for a +3.8 HOTA improvement, yielding a favorable accuracy/latency trade-off for edge deployment.

To characterize dense-association behavior, Table XVI reports a controlled HDA-only profile in which the detector and ReID stages are excluded and the number of active candidates N is varied. The Jacobian/projection term grows approximately linearly with N , whereas the dual-IoU matrix update grows faster because the candidate track-detection matrix expands with traffic density. At $N=400$, the HDA-specific overhead is 8.7 ms/frame under this controlled association-stage profile.

TABLE XVI
CONTROLLED HDA-ONLY LATENCY PROFILE (MS/FRAME). N DENOTES THE NUMBER OF DETECTIONS/ACTIVE CANDIDATES PER FRAME. DETECTOR AND REID STAGES ARE EXCLUDED SO THAT THE TABLE ISOLATES THE ADDITIONAL ASSOCIATION COST OF HDA.

Stage	$N=80$	$N=180$	$N=300$	$N=400$
Jacobian + projection	0.8	1.3	2.0	2.7
Dual-IoU matrix	0.8	1.7	3.1	4.9
Aug. KF pred./upd.	0.3	0.5	0.8	1.1
HDA-specific overhead	1.9	3.5	5.9	8.7

M. Statistical Significance

Each tracker configuration is run 5 times with different random seeds (affecting detector NMS ties and Hungarian tie-breaking); we report mean $\pm$ 95% CI (Student’s t , $n=5$) in Table XVII and perform a paired two-sided t -test between each baseline and its +HDA counterpart. All four HDA

TABLE XVII
STATISTICAL SIGNIFICANCE OF HDA OVER 5 SEEDS. p -VALUES ARE FROM A PAIRED TWO-SIDED t -TEST; ALL IMPROVEMENTS ARE SIGNIFICANT AT $p < 10^{-4}$.

Tracker	HOTA (base)	HOTA (+HDA)	p -value
StrongSORT	40.3 $\pm$ 0.21	44.1 $\pm$ 0.18	1.3×10^{-5}
BoT-SORT-R	41.2 $\pm$ 0.24	45.6 $\pm$ 0.22	7.6×10^{-6}
OC-SORT	40.7 $\pm$ 0.19	44.9 $\pm$ 0.20	2.1×10^{-5}
Deep OC-SORT	41.0 $\pm$ 0.23	45.3 $\pm$ 0.19	1.9×10^{-5}

improvements are highly significant ($p < 10^{-4}$) with 95% CIs that do not overlap with the baseline.

N. Failure-Case Quantification

We categorize remaining errors on the BoT-SORT-R+HDA run: (i) detector misses (38.4% of all track fragments) dominate, mostly for small pedestrians/bikes at the extreme periphery ($r/R_{\max} > 0.9$); (ii) severe occlusions by large vehicles account for 27.1%; (iii) appearance collapse under identical clothing (pedestrian ReID confusion) accounts for 18.6%; (iv) abrupt non-linear motion (sharp U-turns at junctions) accounts for 10.3%; (v) calibration/ISP-induced jitter 5.6%. HDA specifically attacks class (iv); classes (i)–(iii) remain open for future work on dedicated fisheye detectors and ReID models.

Baseline vs. +HDA comparison. The same categorization on the BoT-SORT-R baseline (without HDA) yields: (i) 36.1%, (ii) 25.3%, (iii) 17.4%, (iv) 16.0%, (v) 5.2%. HDA thus reduces the *absolute* fragment count by 23.7% and specifically compresses the non-linear-motion class from 16.0% $\rightarrow$ 10.3% (a 36% relative reduction), confirming that the +4.4 HOTA gain is driven by the geometric error mode rather than by uniform noise reduction.

VII. DISCUSSION

A. Computational Complexity Analysis

A key advantage of HDA is its efficiency compared to full-frame image rectification. Standard dewarping transforms the entire image $I \in \mathbb{R}^{W \times H}$ into a rectified image $I' \in \mathbb{R}^{W' \times H'}$. This process involves a pixel-wise remapping operation with complexity $O(W \times H)$. For high-resolution video (1920×1920), this can add substantial latency and make edge-side real-time processing more difficult.

In contrast, our HDA method operates only on the detected object coordinates. If there are N detections in a frame, the complexity of mapping their bounding boxes is $O(N)$. More precisely, the complete HDA extension (projection, dual-IOU, augmented KF predict) is $O(N + N_T \cdot N_D)$, where N_T and N_D are the numbers of active tracks and detections; the same asymptotic class as the baseline Hungarian step. Since $N \ll W \times H$ (typically $N < 100$), the measured overhead of HDA is limited (9.9% empirically, Sec. VI-L). This design allows the selected real-time configurations to maintain high frame rates while still benefiting from the geometric accuracy of the rectified domain.

Dense-traffic scaling. Table XVI provides a controlled association-stage profile for $N=180$, $N=300$, and $N=400$. The projection/Jacobian term remains smaller than the dual-IOU matrix update, which is the main source of additional cost as the number of candidates increases. These results indicate that dense traffic mainly affects the association matrix stage; stronger gating for center-zone candidates may be considered in future edge deployments.

B. Deployment Challenges

While FE-DETRAC and HDA show promising results, several challenges remain for real-world deployment. First, the fixed camera assumption is crucial for our method; strong wind or pole vibrations can disrupt the calibration. Empirically,

vibration amplitudes of ± 1 pixel at the principal point cause a < 0.3 HOTA drop, matching the noise level explored in Sec. VI-J; amplitudes of ± 5 px (rare, only during storms) cause up to 1.5 HOTA drop, which can be compensated by online re-estimation of the principal point every 30 s. Second, extreme weather conditions like heavy rain on the lens (common in fisheye cameras facing upwards) can obscure the view entirely. Future work will investigate robust calibration maintenance and weather-agnostic detection models.

C. Implications for SD-IOV Multimedia Streaming

The proposed benchmark and tracker are also relevant to software-defined Internet of Vehicles (SD-IOV) deployment. In a smart-city architecture, fisheye video may not always need to be streamed at full resolution once object trajectories are available. Instead, an SD-IOV controller can use the HDA outputs—object trajectories, local density, periphery-zone uncertainty, and congestion-level estimates—as dynamic control signals for bandwidth, edge-computation, and storage allocation. Prior work on SD-IOV automation and orchestration [62] and multimedia-streaming resource allocation [63], [64] indicates that centralized software-defined control can adapt network and computing resources to heterogeneous vehicular demand. FE-DETRAC may provide a perception-driven input to this control loop: when peripheral uncertainty or object density increases, the controller can temporarily allocate higher bitrate to the affected fisheye regions, prioritize low-latency trajectory metadata from roadside units, or shift short-term buffering and ReID computation to a nearby edge server. Conversely, during low-density periods, the system can downlink compact trajectory summaries rather than raw video, reducing backhaul load while preserving traffic-state awareness. This interaction between distortion-aware perception and SD-IOV resource allocation is outside the scope of the present benchmark, but it suggests a possible path for future ITS deployments that jointly consider tracking accuracy, communication latency, and edge-resource utilization.

D. Stability of the Augmented Filter

We present an informal stability argument (not a full convergence proof). Let $\mathbf{P}_t^{(f)}$ be the fisheye-domain error covariance of the baseline tracker and let $\mathbf{P}_t^{(d)}$ be the corrected-domain covariance produced by the Jacobian transport of Sec. V-B. On the admissible FoV $\mathcal{F}$ the Jacobian has bounded operator norm $\|J\|_2 \leq L$, with the conservative radial bound $L = \sec^2(\theta_{\max}) \approx 14.0$ for $\theta_{\max} = 1.3$ rad (Eq. 10), so by the covariance-transport identity

$$\mathbf{P}_t^{(d)} = J \mathbf{P}_t^{(f)} J^\top \preceq L^2 \mathbf{P}_t^{(f)}.$$

Hence, whenever the baseline filter is stable in the standard Kalman sense—i.e., the f -domain pair $(\mathbf{F}, \mathbf{H})$ is detectable and the process/measurement noise are bounded, so that $\mathbf{P}_t^{(f)} \preceq \bar{\mathbf{P}}$ uniformly in t —the augmented covariance $\mathbf{P}_t = \text{diag}(\mathbf{P}_t^{(f)}, \mathbf{P}_t^{(d)})$ is also uniformly bounded by $\text{diag}(\bar{\mathbf{P}}, L^2 \bar{\mathbf{P}})$. Under these bounded-noise assumptions, HDA does not introduce an unbounded corrected-domain covariance

term beyond the baseline covariance bound. The restriction to the admissible FoV $\mathcal{F}$ is crucial—outside $\mathcal{F}$ the map g is unbounded—but in practice every released camera is cropped to $\theta_{\max} \leq 1.3$ rad, so the bound applies throughout our experiments. We deliberately avoid stronger convergence claims (e.g., contraction-rate guarantees) because they require additional assumptions on the baseline's Kalman gain that are implementation-specific.

VIII. CONCLUSIONS

This paper introduced FE-DETRAC, a large-scale infrastructure fisheye video benchmark for urban traffic multi-object tracking. To enable reproducible research, we released 20,000 annotated frames, 470,000 bounding boxes, per-camera intrinsics (Kannala–Brandt model with 0.41-pixel RMS reprojection error), and defined camera-disjoint splits to test generalization.

Addressing the challenge of fisheye distortion, we proposed **Hybrid Data Association (HDA)**, a lightweight, plug-and-play module. HDA couples fisheye-view and corrected-view representations, utilizing a dual-view IoU metric to robustly associate detections even under non-linear motion. Extensive experiments demonstrated that HDA improves the tracking performance (HOTA) of modern trackers like StrongSORT, BoT-SORT-R, OC-SORT, and Deep OC-SORT, with gains of +3.4 to +5.3 HOTA validated under 95% confidence intervals over 5 runs, cross-dataset transfer on FishEye8K, and a full per-zone / per-camera / per-calibration analysis. Together, the benchmark and the proposed HDA module provide a reproducible basis for future research on wide-area traffic monitoring and smart-city perception.

DATA AVAILABILITY STATEMENT

The FE-DETRAC benchmark dataset generated and analyzed during this study is publicly available in the SciDM repository under the DOI: <https://doi.org/10.30193/scidm-ds-571593m>.

ACKNOWLEDGMENT

We thank SciDM and the National Center for High-performance Computing (NCHC) for providing computational and storage resources, and the Emirates Center for Mobility Research (ECMR) through Grant 12R012, United Arab Emirates University (UAEU), Hsinchu City Police Bureau, Hsinchu City Government, and the Central Taiwan Science Park Bureau (NSTC).

REFERENCES

- [1] A. Bewley, Z. Ge, L. Ott, F. Ramos, and B. Upcroft, "Simple online and realtime tracking," in *Proceedings of the IEEE International Conference on Image Processing (ICIP)*, 2016, pp. 3464–3468.
- [2] N. Wojke, A. Bewley, and D. Paulus, "Simple online and realtime tracking with a deep association metric," in *Proceedings of the IEEE International Conference on Image Processing (ICIP)*, 2017, pp. 3645–3649.
- [3] L. Wen, D. Du, Z. Cai, Z. Lei, M.-C. Chang, H. Qi, J. Lim, M.-H. Yang, and S. Lyu, "UA-DETRAC: A new benchmark and protocol for multi-object detection and tracking," *Computer Vision and Image Understanding*, vol. 193, p. 102907, 2020.
- [4] M. Naphade, S. Wang, D. C. Anastasiu, Z. Tang, M.-C. Chang, X. Yang, Y. Yao, L. Zheng, P. Chakraborty, C. E. Lopez *et al.*, "The 5th AI city challenge," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, 2021, pp. 4263–4273.
- [5] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? the KITTI vision benchmark suite," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2012, pp. 3354–3361.
- [6] F. Yu, H. Chen, X. Wang, W. Xian, Y. Chen, F. Liu, V. Madhavan, and T. Darrell, "BDD100K: A diverse driving dataset for heterogeneous multitask learning," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 2633–2642.
- [7] S. Yogamani, C. Hughes, J. Horgan, G. Sistu, P. Varley, D. O'Dea, M. Uricár, S. Milz, M. Simon, K. Amende *et al.*, "WoodScape: A multi-task, multi-camera fisheye dataset for autonomous driving," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 9308–9318.
- [8] X. Wang, X. Zhang, Y. Zhu, Y. Guo, X. Yuan, L. Xiang, Z. Wang, G. Ding, D. J. Brady, Q. Dai, and L. Fang, "PANDA: A gigapixel-level human-centric video dataset," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 3265–3275.
- [9] M. Gochoo, M.-E. Otgonbold, E. Ganbold, J.-W. Hsieh, M.-C. Chang, P.-Y. Chen, B. Dorj, H. Al Jassmi, G. Batnasan, F. Alnajjar, M. Abduljabbar, and F.-P. Lin, "FishEye8K: A benchmark and dataset for fisheye camera object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, June 2023, pp. 5305–5313.
- [10] M. Everingham, L. Van Gool, C. K. Williams, J. Winn, and A. Zisserman, "The Pascal visual object classes (VOC) challenge," *International Journal of Computer Vision*, vol. 88, no. 2, pp. 303–338, 2010.
- [11] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft COCO: Common objects in context," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2014, pp. 740–755.
- [12] K. Bernardin and R. Stiefelhausen, "Evaluating multiple object tracking performance: The CLEAR MOT metrics," *EURASIP Journal on Image and Video Processing*, vol. 2008, pp. 1–10, 2008.
- [13] J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," *arXiv preprint arXiv:1804.02767*, 2018.
- [14] A. Wang, T. Qiu, and L. Shao, "A simple method of radial distortion correction with centre of distortion estimation," *Journal of Mathematical Imaging and Vision*, vol. 35, pp. 165–172, 2009.
- [15] Y. Du, Z. Zhao, Y. Song, Y. Zhao, F. Su, T. Gong, and H. Meng, "StrongSORT: Make DeepSORT great again," *IEEE Transactions on Multimedia*, vol. 25, pp. 8725–8737, 2023.
- [16] N. Aharon, R. Orfaig, and B.-Z. Bobrovsky, "BoT-SORT: Robust associations multi-pedestrian tracking," *arXiv preprint arXiv:2206.14651*, 2022.
- [17] J. Cao, J. Pang, X. Weng, R. Khirrodar, and K. Kitani, "Observation-centric SORT: Rethinking SORT for robust multi-object tracking," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 9686–9696.
- [18] G. Maggolino, A. Ahmad, J. Cao, and K. Kitani, "Deep OC-SORT: Multi-pedestrian tracking by adaptive re-identification," in *Proceedings of the IEEE International Conference on Image Processing (ICIP)*, 2023, pp. 3025–3029.
- [19] A. Bochkovskiy, C.-Y. Wang, and H.-Y. M. Liao, "YOLOv4: Optimal speed and accuracy of object detection," *arXiv preprint arXiv:2004.10934*, 2020.
- [20] G. Jocher *et al.*, "YOLOv5," 2021. [Online]. Available: <https://github.com/ultralytics/yolov5>
- [21] C.-Y. Wang, A. Bochkovskiy, and H.-Y. M. Liao, "YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors," *arXiv preprint arXiv:2207.02696*, 2022.
- [22] C. Papageorgiou and T. Poggio, "A trainable system for object detection," *International Journal of Computer Vision*, vol. 38, no. 1, pp. 15–33, 2000.
- [23] Z. Luo, F. B.-Charron, C. Lemaire, J. Konrad, S. Li, A. Mishra, A. Achkar, J. Eichel, and P.-M. Jodoin, "MIO-TCO: A new benchmark dataset for vehicle classification and localization," *IEEE Transactions on Image Processing*, vol. 27, no. 10, pp. 5129–5141, 2018.
- [24] M.-R. Hsieh, Y.-L. Lin, and W. H. Hsu, "Drone-based object counting by spatially regularized regional proposal network," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 4145–4153.

- [25] D. Du, Y. Qi, H. Yu, Y. Yang, K. Duan, G. Li, W. Zhang, Q. Huang, and Q. Tian, "The unmanned aerial vehicle benchmark: Object detection and tracking," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 370–386.
- [26] P. Zhu, L. Wen, D. Du, X. Bian, H. Fan, Q. Hu, and H. Ling, "VisDrone-DET2018: The vision meets drone object detection in image challenge results," in *Proceedings of the European Conference on Computer Vision (ECCV) Workshops*, 2018, pp. 437–468.
- [27] H. Rashed, E. Mohamed, G. Sistu, V. R. Kumar, C. Eising, A. El-Sallab, and S. Yogamani, "Generalized object detection on fisheye cameras for autonomous driving: Dataset, representations and baseline," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2021, pp. 2272–2280.
- [28] D. Zhang, T. Yang, and B. Zhao, "Swin-fisheye: Object detection for fisheye images," *IET Image Processing*, vol. 18, no. 13, pp. 3904–3915, 2024.
- [29] R. Kvyetnyy, O. Sofina, M. Adamski, and O. Romanchuk, "Object detection in images with low light condition," in *Proceedings of Photonics Applications in Astronomy, Communications, Industry, and High-Energy Physics Experiments*, vol. 10445. SPIE, 2017, p. 104450W.
- [30] I. Klein, "NEXET: The largest and most diverse road dataset in the world," Nexar Blog, August 2017. [Online]. Available: <https://www.nexar-ai.com/blog/nexet-the-largest-and-most-diverse-road-dataset-in-the-world>
- [31] C. H. Bahnsen and T. B. Moeslund, "Rain removal in traffic surveillance: Does it matter?" *IEEE Transactions on Intelligent Transportation Systems*, vol. 20, no. 8, pp. 2802–2819, 2019.
- [32] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich feature hierarchies for accurate object detection and semantic segmentation," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2014, pp. 580–587.
- [33] R. Girshick, "Fast R-CNN," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2015, pp. 1440–1448.
- [34] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 39, no. 6, pp. 1137–1149, 2017.
- [35] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "SSD: Single shot multibox detector," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2016, pp. 21–37.
- [36] J. Redmon, "Darknet: Open source neural networks in C," 2013. [Online]. Available: <http://pjreddie.com/darknet/>
- [37] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, and S. Belongie, "Feature pyramid networks for object detection," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2117–2125.
- [38] H. Zhang, M. Cisse, Y. N. Dauphin, and D. Lopez-Paz, "mixup: Beyond empirical risk minimization," in *Proceedings of the International Conference on Learning Representations (ICLR)*, 2018.
- [39] S. Yun, D. Han, S. J. Oh, S. Chun, J. Choe, and Y. Yoo, "CutMix: Regularization strategy to train strong classifiers with localizable features," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 6023–6032.
- [40] C.-Y. Wang, I.-H. Yeh, and H.-Y. M. Liao, "You only learn one representation: Unified network for multiple tasks," *arXiv preprint arXiv:2105.04206*, 2021.
- [41] —, "YOLOv9: Learning what you want to learn using programmable gradient information," *arXiv preprint arXiv:2402.13616*, 2024.
- [42] A. Wang, H. Chen, L. Liu, K. Chen, Z. Lin, J. Han, and G. Ding, "YOLOv10: Real-time end-to-end object detection," *arXiv preprint arXiv:2405.14458*, 2024.
- [43] Y. Zhao, W. Lv, S. Xu, J. Wei, G. Wang, Q. Dang, Y. Liu, and J. Chen, "DETRs beat YOLOs on real-time object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 16 965–16 974.
- [44] K. Duan, S. Bai, L. Xie, H. Qi, Q. Huang, and Q. Tian, "CenterNet: Keypoint triplets for object detection," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 6569–6578.
- [45] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2017, pp. 2980–2988.
- [46] R. E. Kalman, "A new approach to linear filtering and prediction problems," *Transactions of the ASME—Journal of Basic Engineering*, vol. 82, no. 1, pp. 35–45, 1960.
- [47] Z. Wang, L. Zheng, Y. Liu, Y. Li, and S. Wang, "Towards real-time multi-object tracking," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2020, pp. 107–122.
- [48] Y. Zhang, C. Wang, X. Wang, W. Zeng, and W. Liu, "FairMOT: On the fairness of detection and re-identification in multiple object tracking," *International Journal of Computer Vision*, vol. 129, pp. 3069–3087, 2021.
- [49] P. Sun, J. Cao, Y. Jiang, R. Zhang, E. Xie, Z. Yuan, C. Wang, and P. Luo, "TransTrack: Multiple object tracking with transformer," *arXiv preprint arXiv:2012.15460*, 2020.
- [50] Y. Zhang, P. Sun, Y. Jiang, D. Yu, F. Weng, Z. Yuan, P. Luo, W. Liu, and X. Wang, "ByteTrack: Multi-object tracking by associating every detection box," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2022, pp. 1–21.
- [51] P. Vanasirikul, P. Charoenpoonpanich, and E. Chuangsuwanich, "Distortion-aware multi-object tracking via virtual plane projection in overhead fisheye cameras," in *Proceedings of the British Machine Vision Conference (BMVC)*. BMVA, 2025. [Online]. Available: https://bmva-archive.org.uk/bmvc/2025/assets/papers/Paper_880/paper.pdf
- [52] C.-I. Huang and J.-W. Hsieh, "FE-DETRAC: Infrastructure fisheye video benchmark with distortion-aware hybrid data association for multi-object tracking," 2026. [Online]. Available: <https://doi.org/10.30193/scidm-ds-571593m>
- [53] J. Cohen, "A coefficient of agreement for nominal scales," *Educational and Psychological Measurement*, vol. 20, no. 1, pp. 37–46, 1960.
- [54] D. Scaramuzza, A. Martinelli, and R. Siegwart, "A toolbox for easily calibrating omnidirectional cameras," in *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2006, pp. 5695–5701.
- [55] J. Kannala and S. S. Brandt, "A generic camera model and calibration method for conventional, wide-angle, and fish-eye lenses," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 28, no. 8, pp. 1335–1340, 2006.
- [56] Z. Zhang, "A flexible new technique for camera calibration," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 22, no. 11, pp. 1330–1334, 2000.
- [57] A. Goodarzi, E. K. Stathopoulou, and F. Remondino, "Geometric calibration of full spherical panoramic ricoh-theta camera," in *ISPRS Annals of Photogrammetry, Remote Sensing and Spatial Information Sciences*, 2017, pp. 237–244.
- [58] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.
- [59] Y. Ma, D. Yu, T. Wu, and H. Wang, "PaddlePaddle: An open-source deep learning platform from industrial practice," *Frontiers of Data and Computing*, vol. 1, no. 1, pp. 105–115, 2019.
- [60] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.
- [61] P.-Y. Chen, M.-C. Chang, J.-W. Hsieh, and Y.-S. Chen, "Parallel residual bi-fusion feature pyramid network for accurate single-shot object detection," *IEEE Transactions on Image Processing*, vol. 30, pp. 9099–9111, 2021.
- [62] S. R. Pokhrel, "Software defined internet of vehicles for automation and orchestration," *IEEE Transactions on Intelligent Transportation Systems*, vol. 22, no. 6, pp. 3890–3899, 2021.
- [63] A. Montazerolghaem, "Efficient resource allocation for multimedia streaming in software-defined internet of vehicles," *IEEE Transactions on Intelligent Transportation Systems*, vol. 24, no. 12, pp. 14 718–14 731, 2023.
- [64] X. Jiang, F. R. Yu, T. Song, and V. C. M. Leung, "Resource allocation of video streaming over vehicular networks: A survey, some research issues and challenges," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 7, pp. 5955–5975, 2022.